

Deep Learning — Final Exam

Date: Thursday, June 15, 2023

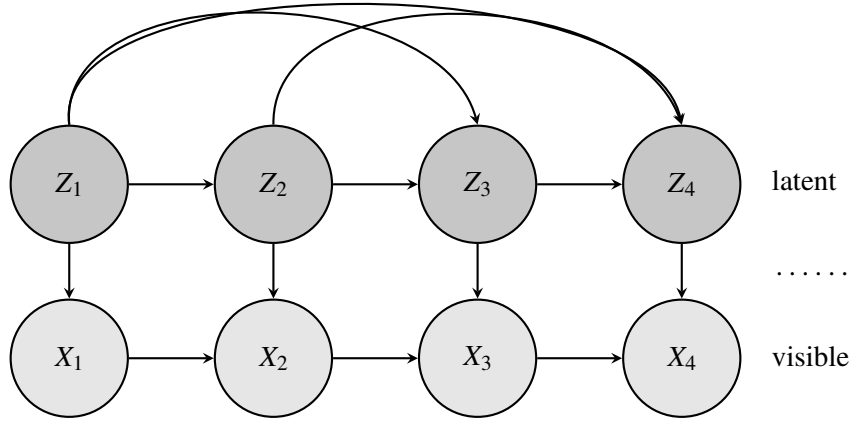
Time: 12:20pm – 15:10pm (170 minutes)

Format: Open book

Instructions:

- 1) You may give your answers in Chinese or English.
- 2) Please give your answers in succinct phrases or point form.
- 3) Please write your answers clearly (with explicit denotation of labels and symbols used).

1. (25 pts) Given the following latent factor model, design a variational autoencoder to learn the distribution $p(X_1, X_2, \dots, X_T)$.



- (a) (3 pts) According to the graphical model, factorize the encoding distribution $q(Z_{1-T}|X_{1-T}) = \prod_{t=1}^T q(Z_t|\dots)$. Apply the d-separation rule to identify the conditioning factors in “ $\dots$ ”.
- (b) (3 pts) By observing $q(Z_t|\dots)$, choose a neural network to implement it. You must specify (a) the network type, e.g. CNN or RNN, and the rationale; (b) indicate explicitly its input and output, and whether there is any hidden connection and/or output recurrence; and (c) specify the assumption about the distribution.
- (c) (3 pts) According to the graphical model, factorize the decoding distribution $p(X_{1-T}|Z_{1-T}) = \prod_{t=1}^T p(X_t|\dots)$. Apply the d-separation rule to identify the conditioning factors in “ $\dots$ ”.
- (d) (3 pts) By observing $p(X_t|\dots)$, choose a neural network to implement it. You must specify (a) the network type, e.g. CNN or RNN, and the rationale; (b) indicate explicitly its input and output, and whether there is any hidden connection and/or output recurrence; and (c) specify the assumption about the distribution.
- (e) (3 pts) Choose a neural network to implement the prior distribution $P(Z_1, Z_2, \dots, Z_T)$. You must specify (a) the network type, e.g. CNN or RNN, and the rationale; (b) indicate explicitly its input and output, and whether there is any hidden connection and/or output recurrence; and (c) specify the assumption about the distribution.

- (f) (10 pts) Draw a diagram to connect all these networks together for end-to-end training and specify explicitly the loss function. If re-parameterization is needed, indicate where it should be used.
2. (25 pts) Please compare the following generative models: (1) the RNN-based auto-regressive model, (2) variational autoencoder (VAE), (3) generative adversarial network (GAN), (4) flow model, and (5) diffusion model.
- (a) (5 pts) Which models would allow an exact evaluation of the likelihood of a data sample?
- (b) (5 pts) Specify the training objective for each of these models, e.g. maximizing the data likelihood, evidence lower bound, or JS-divergence.
- (c) (5 pts) In terms of the data generation process, please describe how to draw samples from each of these models. Be specific about whether the generator generates data samples or data distributions.
- (d) (3 pts) What are the major advantages of the diffusion model as compared to GAN?
- (e) (4 pts) In designing a diffusion model, a very small Gaussian noise is introduced in every forward diffusion step. That is, $x_t = \sqrt{1 - \beta_t}x_{t-1} + N$, where N is a Gaussian noise with mean zero and a small variance β_t . Please explain why β_t has to be small in order to ensure that the inverse denoising process $p_\theta(x_{t-1}|x_t)$ follows a conditional Gaussian distribution.
- (f) (3 pts) What are the major differences between GAN and W-GAN in terms of their training objectives? Does the critic in W-GAN play the same role as the discriminator in GAN?
3. (6 pts) What techniques are used for exploration in DQN and DDPG respectively?
4. (15 pts) For the following questions, please answer True or False, and explain.
- (a) (3 pts) DQN is an on-policy algorithm.
- (b) (3 pts) DQN learn the Q-function of the current policy.
- (c) (3 pts) Experience replay improves the sample efficiency in DQN.
- (d) (3 pts) The purpose of the Target Network is to provide a stable target for updating the Q-values.
- (e) (3 pts) In general, the target network is updated more frequently than the behavior network.
5. (4 pts) What can be used to reduce variance for the REINFORCE algorithm?

6. (5 pts) What is the Transformer architecture and how does it differ from traditional recurrent neural networks (RNNs)? Provide an example of a task where the Transformer architecture has been successfully applied.
7. (5 pts) Explain the idea of dropout and its role in regularizing neural networks. How does it differ from other regularization techniques such as L1 and L2 regularization? Provide an example of how dropout can be applied in a deep learning model.
8. (5 pts) Explain how the choice of padding affects the size of the output feature map in a convolutional layer. Provide an example to illustrate your explanation.
9. (5 pts) Explain in detail how an attention mechanism works within a neural network. Provide specific examples of how it can improve the performance of a neural network and discuss the underlying principles that enable it to selectively focus on certain parts of the input data. Additionally, compare and contrast the use of attention mechanisms in different types of neural networks such as CNNs and RNNs.
10. (5 pts) Please explain what the sigmoid function and ReLU are. How does the sigmoid function cause the vanishing gradient problem in neural networks? How is it different from ReLU?